

Introducing GIS

week 2nd

Objectives:	Understanding what GIS is and what it can be used for.
Keywords:	GIS, Computer, Maps, Data, Information System, Spatial, Analysis

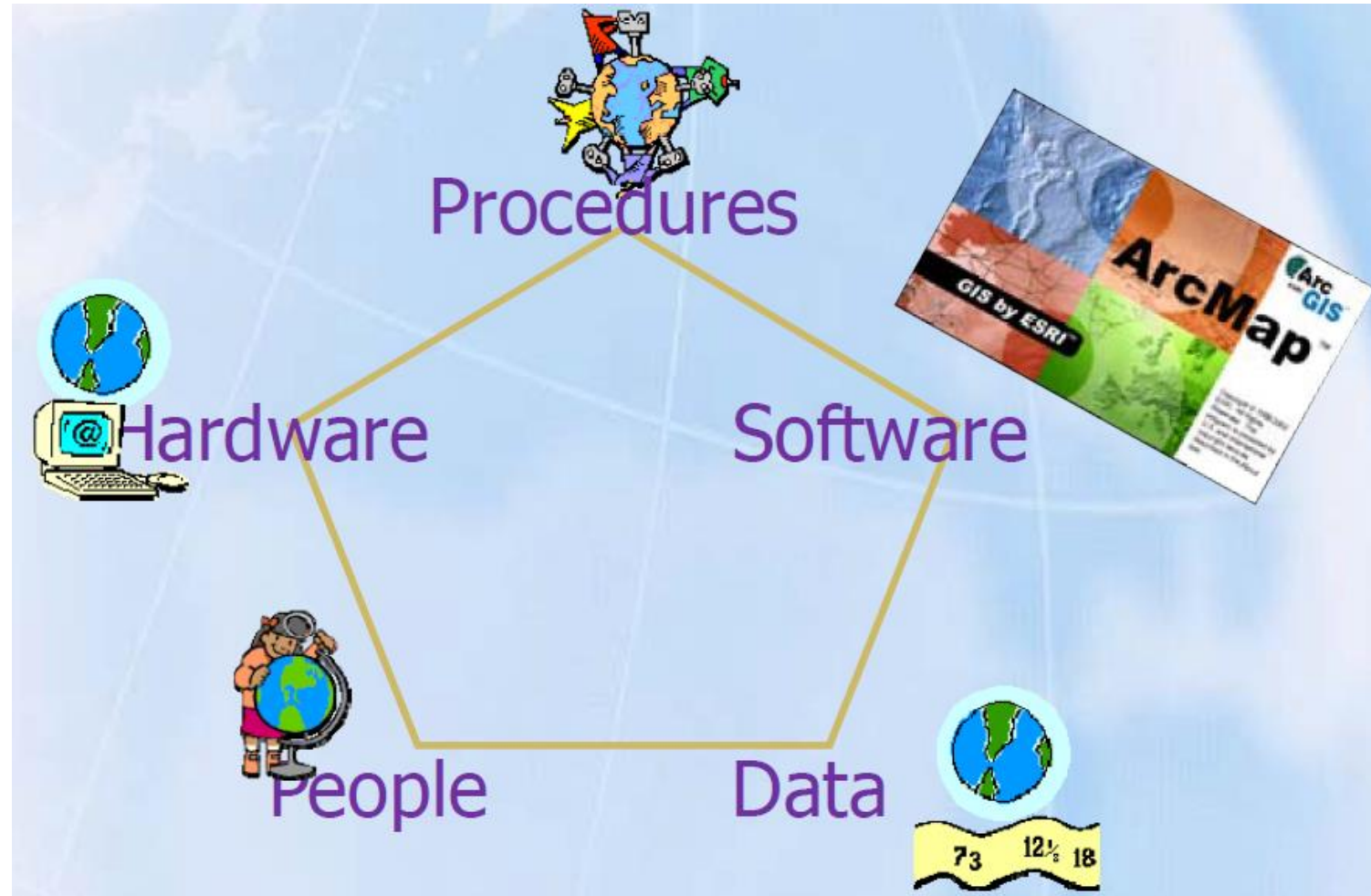
History of GIS

- **First GIS – CGIS (Beginning of 1960s) Canada Geographic System.** Its initial task was to classify and map the land resources of Canada.
 - Canadian Department of Forestry and Rural Development.
 - Purpose: to collect and analyze Canada Land Inventory
 - Still operating today
 - Its development provided many conceptual and technical contributions
- **Environmental Systems Research Institute (ESRI) was founded in 1969 by Jack Dangermond**
 - Utilized CGIS ideas with release of Arc/INFO in the '80s
 - Initially application was in forestry
 - Currently constitutes 50+ % market share

History of GIS contd...

- **Constant feedback from industry and users has led to rapid evolution of GIS:**
 - Improvement in software tools and an increase in software availability
 - Improvements in computing speed and memory
 - Increased quantities and qualities of data (i.e. remote sensing)
 - Improvements in analytical and statistical techniques
- **There are now scores of free GIS software packages available online Today, GIS can:**
 - provide driving directions
 - map criminal behavior
 - monitor voting trends
 - evaluate land use changes and geo-hazards
 - provide community planners with info on development

Components of GIS



Components of GIS

HARDWARE

(PCs, Printers, Plotters, Scanners, Digitizing Tablets, Networks, Storage Devices etc.)

SOFTWARE

(MapInfo, AutoCAD Map, Arcview, ArcGIS, QGIS, IDRISI etc.)

PEOPLE

(Specialists, Developers, Analysts, Users, Viewers)

DATA

(spatial, aspatial; temporal, maps, images, reports, survey points etc.)

APPLICATIONS

(methods, procedures, customized interfaces, GUI, atlases etc.)

Capabilities / What can a GIS do ?

The questions that a GIS is required to answer are mainly:

What is at.....?

Where is it.....?

How has it changed.....?

Which data are related?

What if.....?

LOCATION – What is at ??

- *Here we are seeking to find out what exists at a particular location. A location can be described in many different ways using, for example, place name, zip code, or latitude and longitude coordinates.*



**Who owns the lot at 3233 Texas Street,
and what is it zoned for?**

Identifier:	565-88-221
Area:	108,900,245
Owner:	John Morris
Address:	3233 Texas St.
Zoned land use:	Industrial
Assessment:	\$950,000

CONDITION -- Where is it?

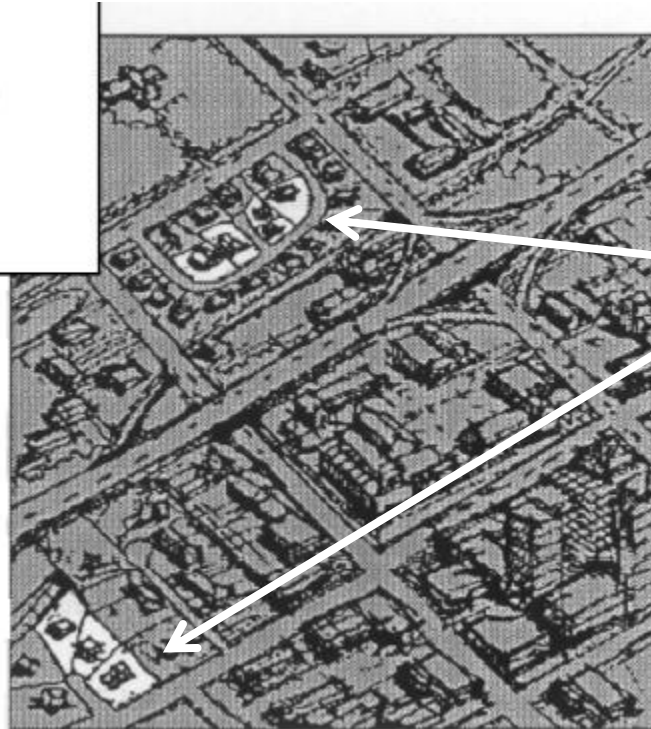
- *Instead of identifying what exists at a given location, you want to find a location where certain conditions are satisfied. For example, you wish to find a house assessed at less than \$20,000 with 4 bedrooms and made of wood or fiber.*

Residential land use

Assessed at less than \$200,000

4 bedrooms

Made of wood or stucco



Where are houses located that you might consider buying?

TRENDS -- How has it changed.....?



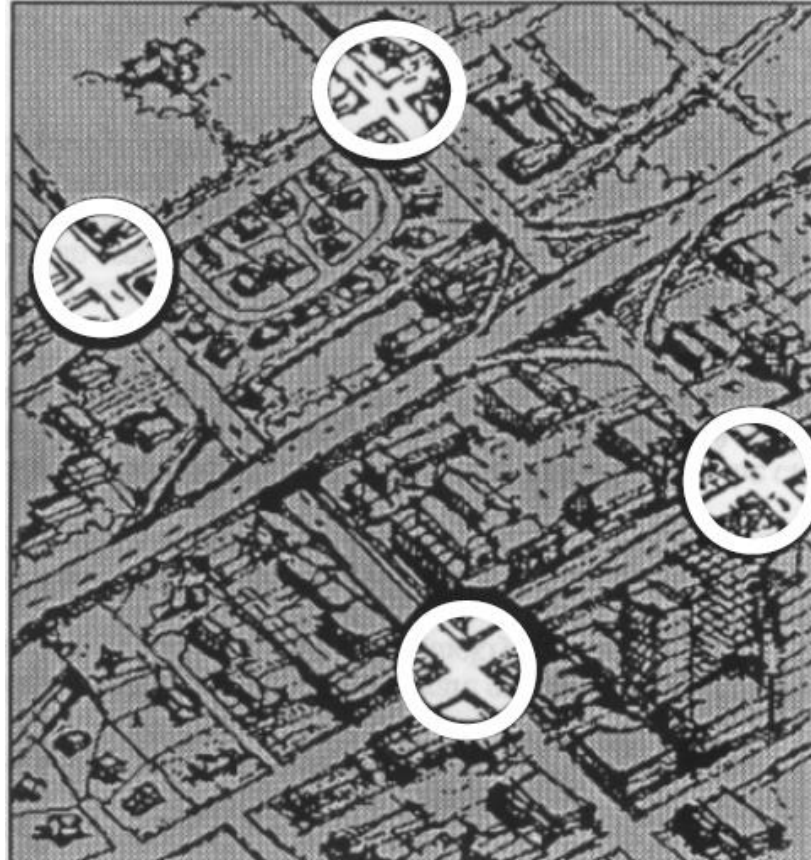
Smallville in 1950



Smallville in 1990

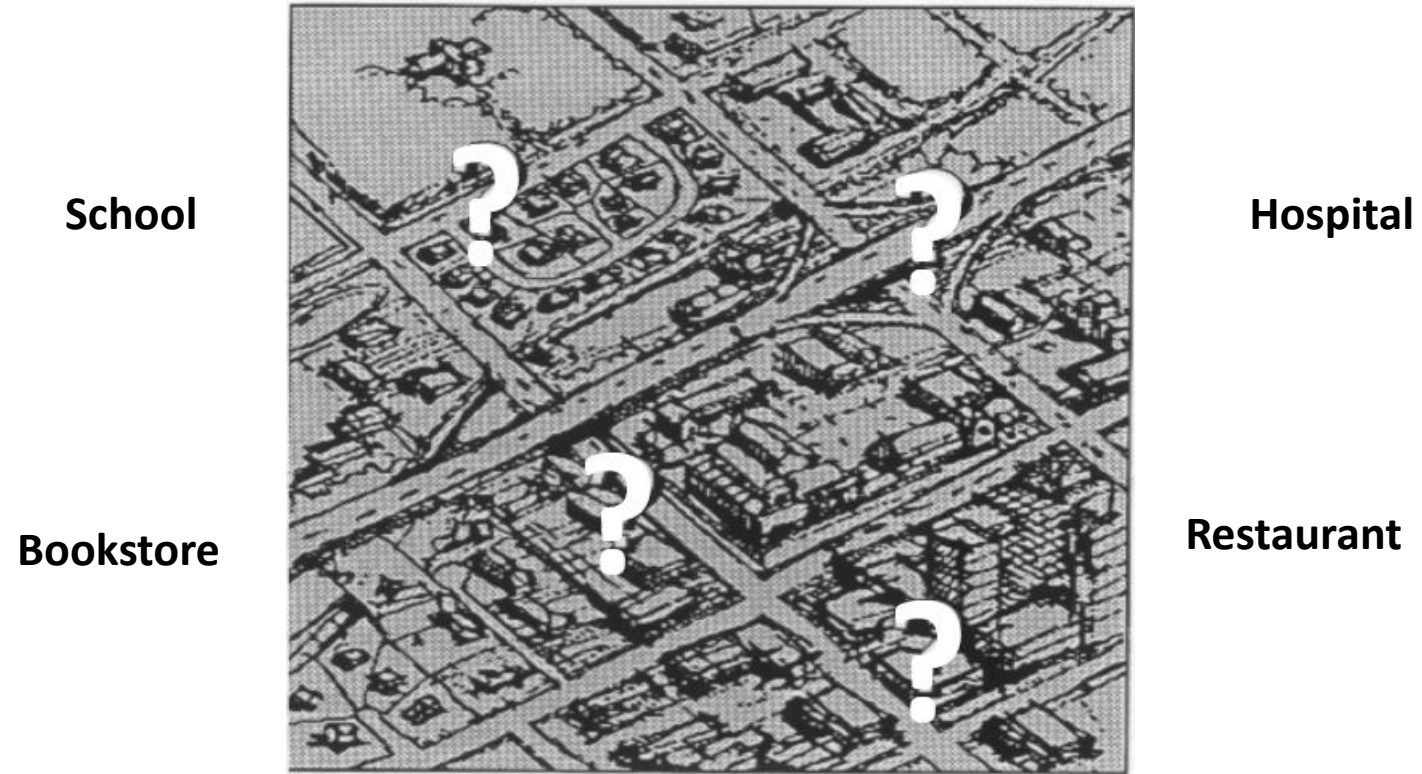
How much land has gone from agriculture to other uses since 1950?

PATTERNS--- Which data are related...?



**What kinds of patterns exist for motor vehicle accidents?
Where do they occur and at what times?**

MODELS -- What If.....?



If you wanted to open a new facility, where would you locate it?

Overview

- Just as we use a word processor to write documents and deal with words on a computer,
- we can use a **GIS application** to deal with **spatial information** on a computer.
- GIS stands for '**Geographical Information System**'.

Overview

- A GIS consists of:
- **Digital Data** — the geographical information that you will view and analyze using computer hardware and software.
- **Computer Hardware** — computers used for storing data, displaying graphics and processing data.
- **Computer Software** — computer programs that run on the computer hardware and allow you to work with digital data. A software program that forms part of the GIS is called a GIS Application

With a GIS application you can :

- open digital maps on your computer,
- create new spatial information to add to a map,
- create printed maps customized to your needs and
- perform spatial analysis.

Let's look at a little example of how GIS can be useful.

- Imagine you are a health worker and you make a note of the date and place of residence of every patient you treat.

Longitude	Latitude	Disease	Date
26.870436	-31.909519	Mumps	13/12/2008
26.868682	-31.909259	Mumps	24/12/2008
26.867707	-31.910494	Mumps	22/01/2009
26.854908	-31.920759	Measles	11/01/2009
26.855817	-31.921929	Measles	26/01/2009
26.852764	-31.921929	Measles	10/02/2009
26.854778	-31.925112	Measles	22/02/2009
26.869072	-31.911988	Mumps	02/02/2009
26.863354	-31.916406	Chicken Pox	26/02/2009

If you look at the table above you will quickly see that there were a lot of measles cases in January and February.

- Our health worker recorded the location of each patient's house by noting its latitude and longitude in the table.
- Using this data in a GIS Application, we can quickly understand a lot more about the patterns of illness:

Example

- *Showing disease records in a GIS application.*
- *It is easy to see that the mumps patients all live close to each other.*

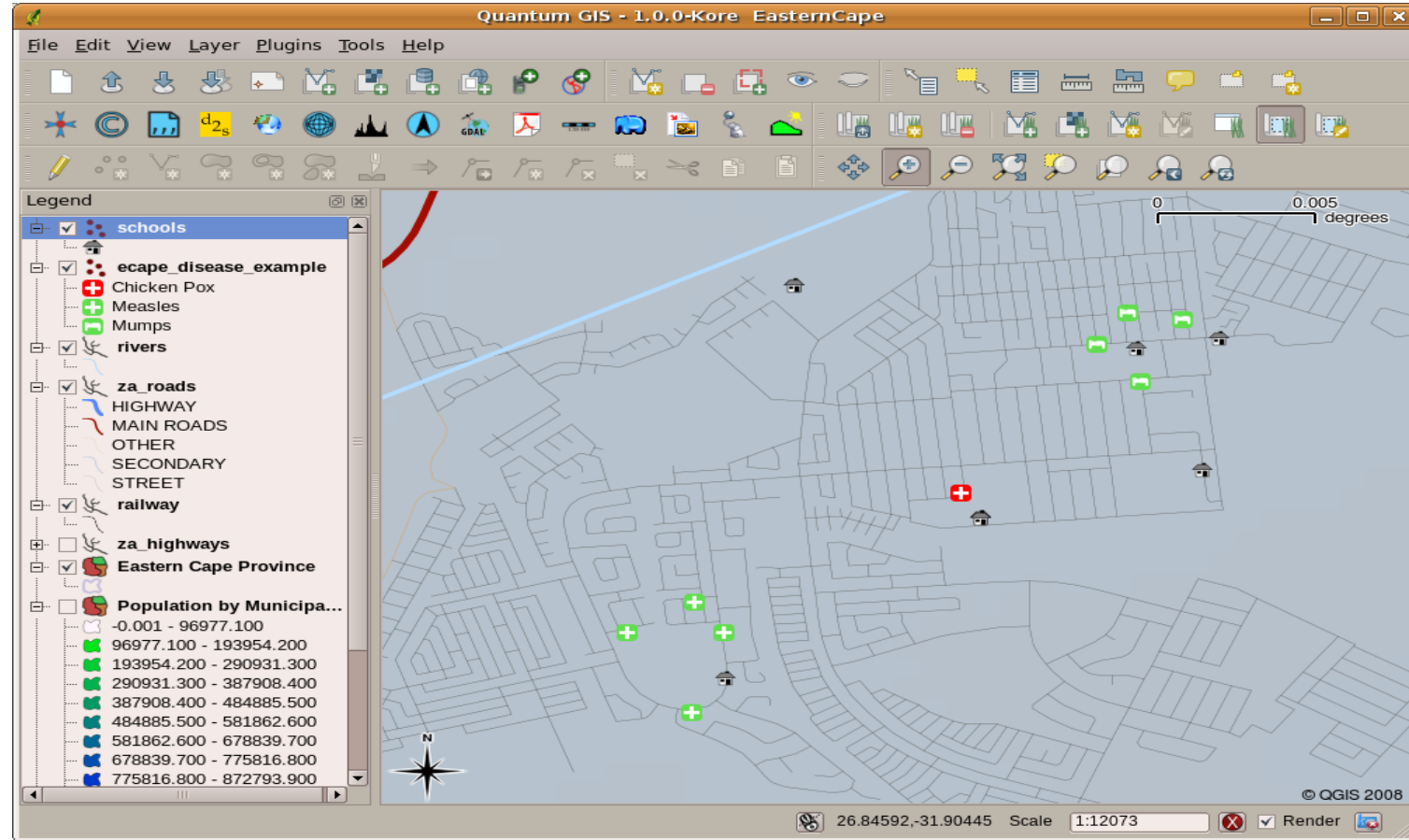


Fig. 1

More about GIS

- GIS is a relatively new field — it started in the 1970's.
- It used to be that computerized GIS was only available to companies and universities that had expensive computer equipment.
- These days, anyone with a personal computer or laptop can use GIS software.
- Over time GIS Applications have also become easier to use — it used to require a lot of training to use a GIS Application, but now it is much easier to get started in GIS even for amateurs and casual users.
- it refers to all aspects of managing and using digital geographical data..

What is GIS Software / a GIS Application?

- You can see an example of what a **GIS Application** looks like Fig.1
- GIS Applications are normally programs with a graphical user interface that can be manipulated using the mouse and keyboard.
- The application provides **menus** near to the top of the window (File, Edit etc.) which, when clicked using the mouse, show a panel of **actions**.
- These actions provide a way for you to tell the GIS Application what you want to do.
- **For example** you may use the menus to tell the GIS Application to add a new layer to the display output.

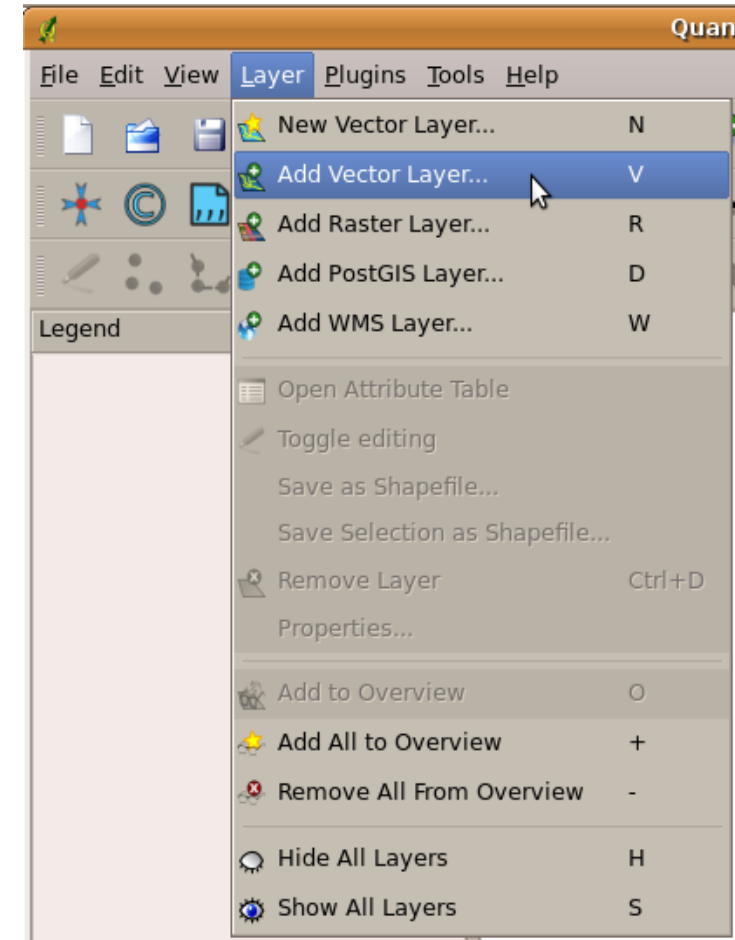


Fig. 2. Application menus, when clicked with the mouse, expand to show a list of actions that can be carried out.

Menus near to the top of the window (File, Edit etc.) which, when clicked using the mouse, show a panel of **actions**.

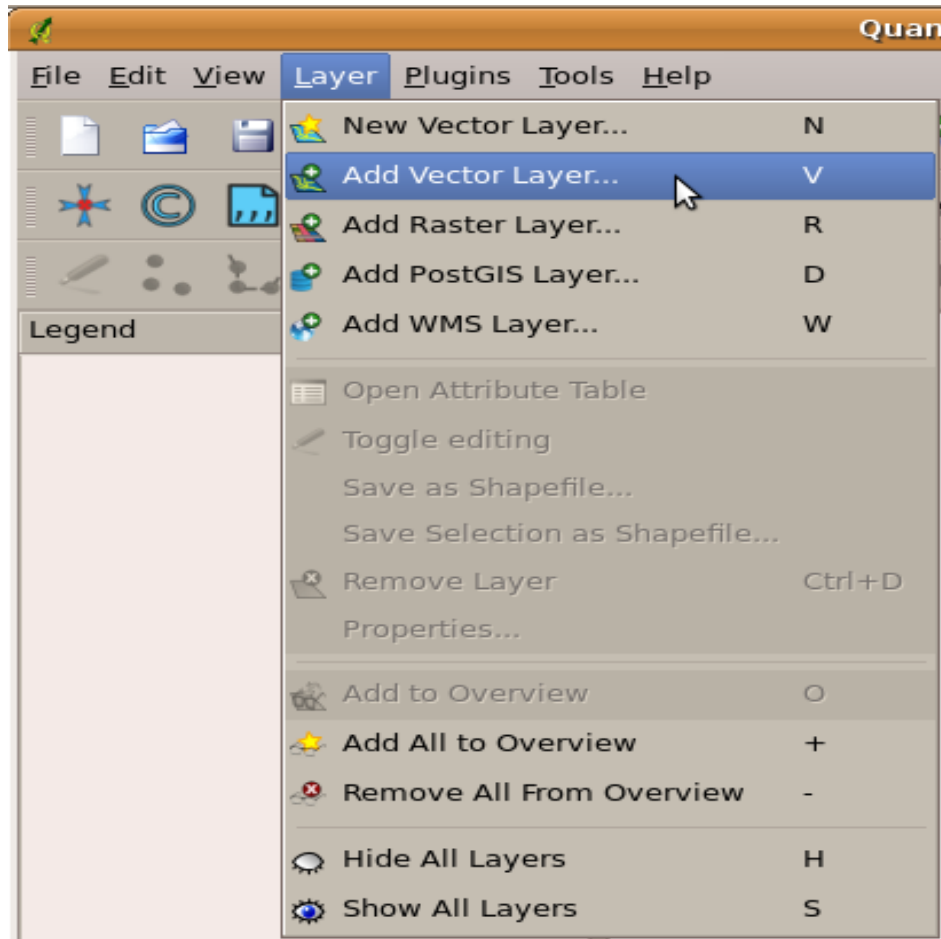


Fig. 2. *Application menus, when clicked with the mouse, expand to show a list of actions that can be carried out.*

Toolbars

- (rows of small pictures that can be clicked with the mouse)
- normally sit just below the menus and provide a quicker way to use frequently needed actions.



Fig. 3. *Toolbars provide quick access to commonly used functions. Holding your mouse over a picture will usually tell you what will happen when you click on it*

Map layers & View

Map layers

- A common function of GIS Applications is to display **map layers**.
- Map layers are stored as files on a disk or as records in a database.
- Normally each map layer will represent something in the real world — a roads layer for example will have data about the street network.

Map View

- When you open a layer in the GIS Application it will appear in the **map view**.
- The map view shows a graphic representing your layer.
- When you add more than one layer to a map view, the layers are overlaid on top of each other.
- Look at figures **Fig.3.**, **Fig.4**, **Fig.5** and **Fig.6** to see a map view that has several layers being added to it.
- An important function of the map view is to allow you to **zoom in to magnify**, **zoom out to see a greater area** and **move around** (panning) in the map.

Fig. 3. A map view with several layers being added to it. A towns layer added to the map view.

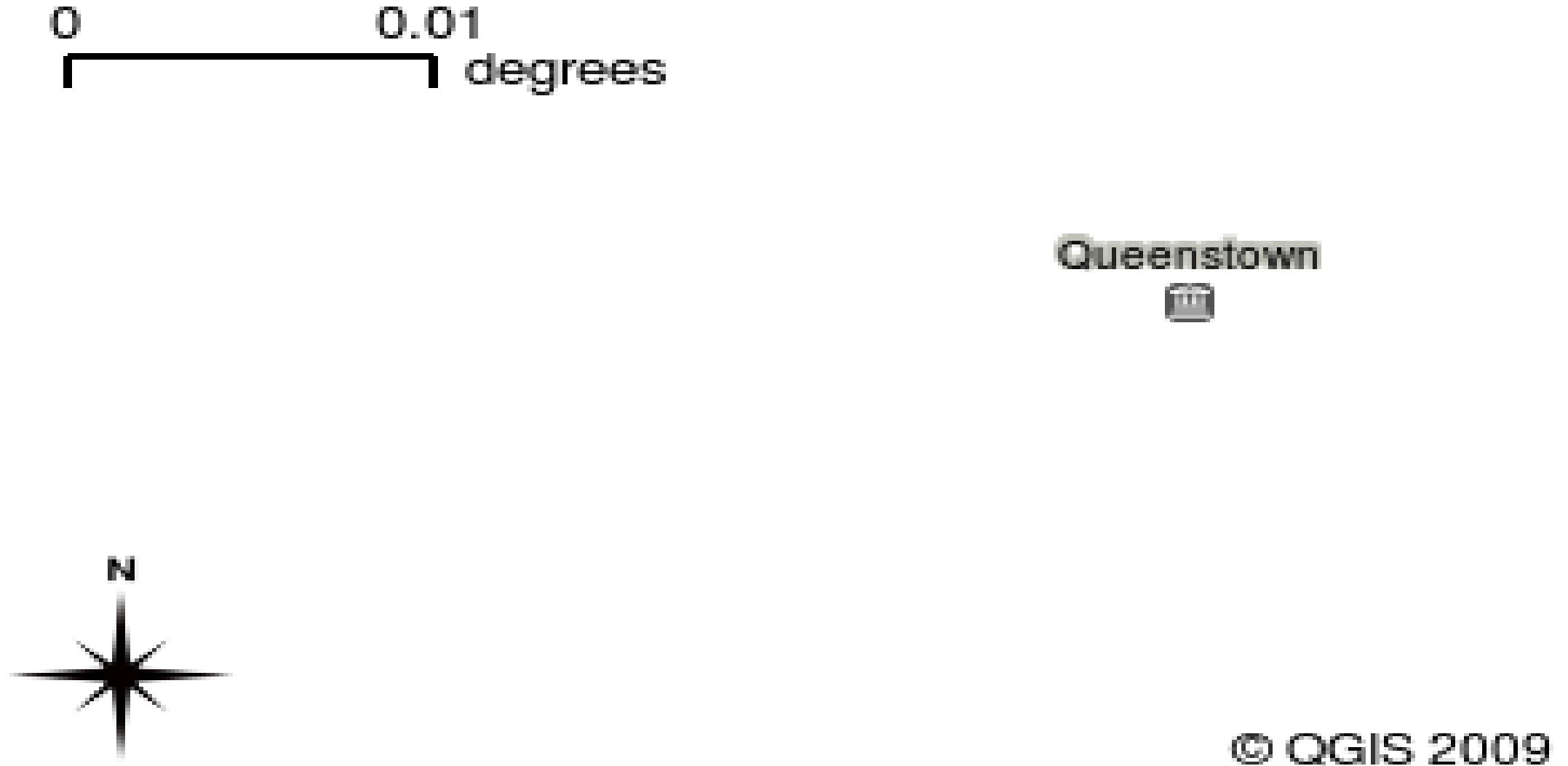


Fig. 4 A map view with several layers being added to it. A schools layer added to the map view

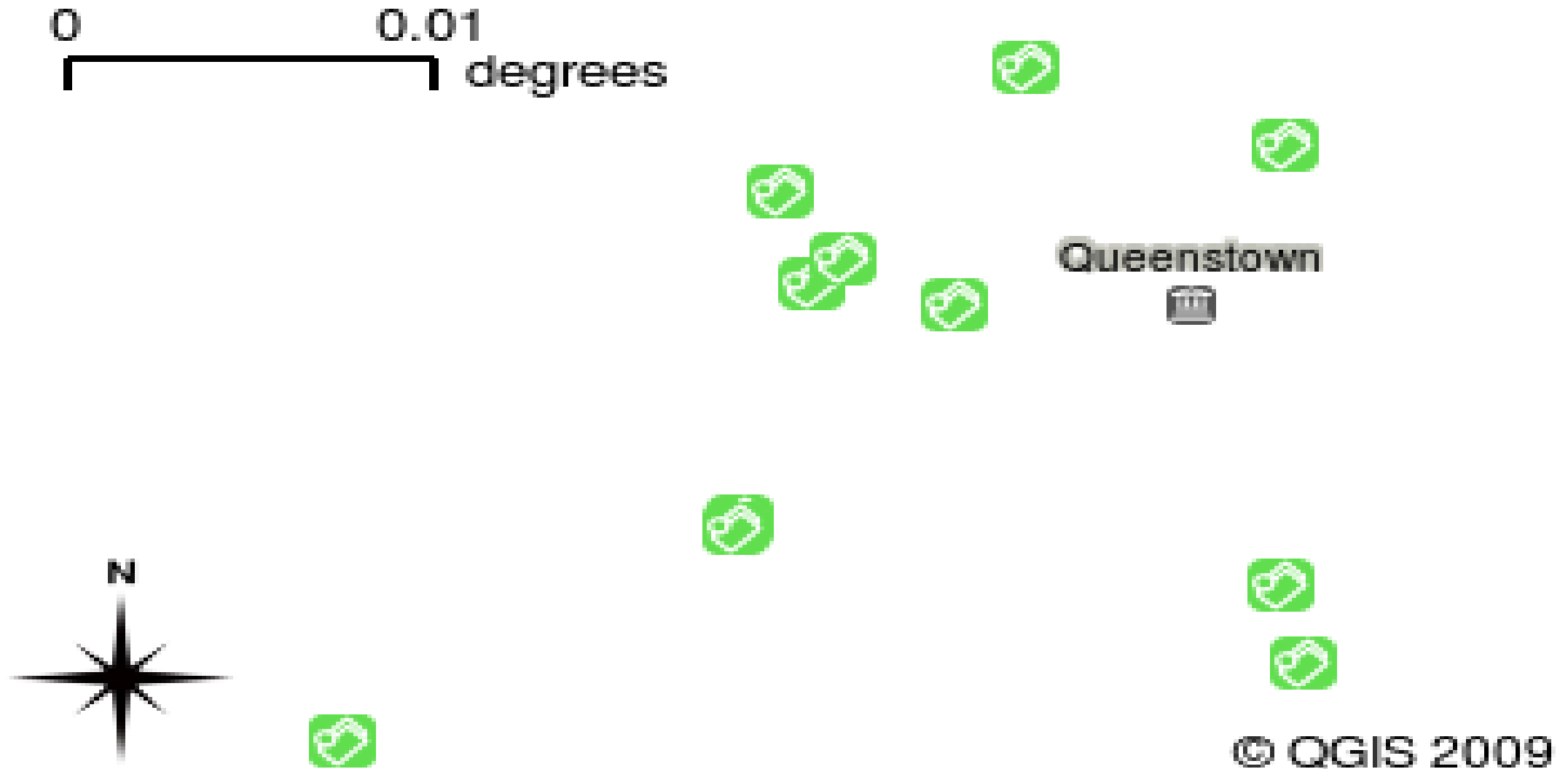


Fig. 5 A map view with several layers being added to it. A railways layer added to the map view

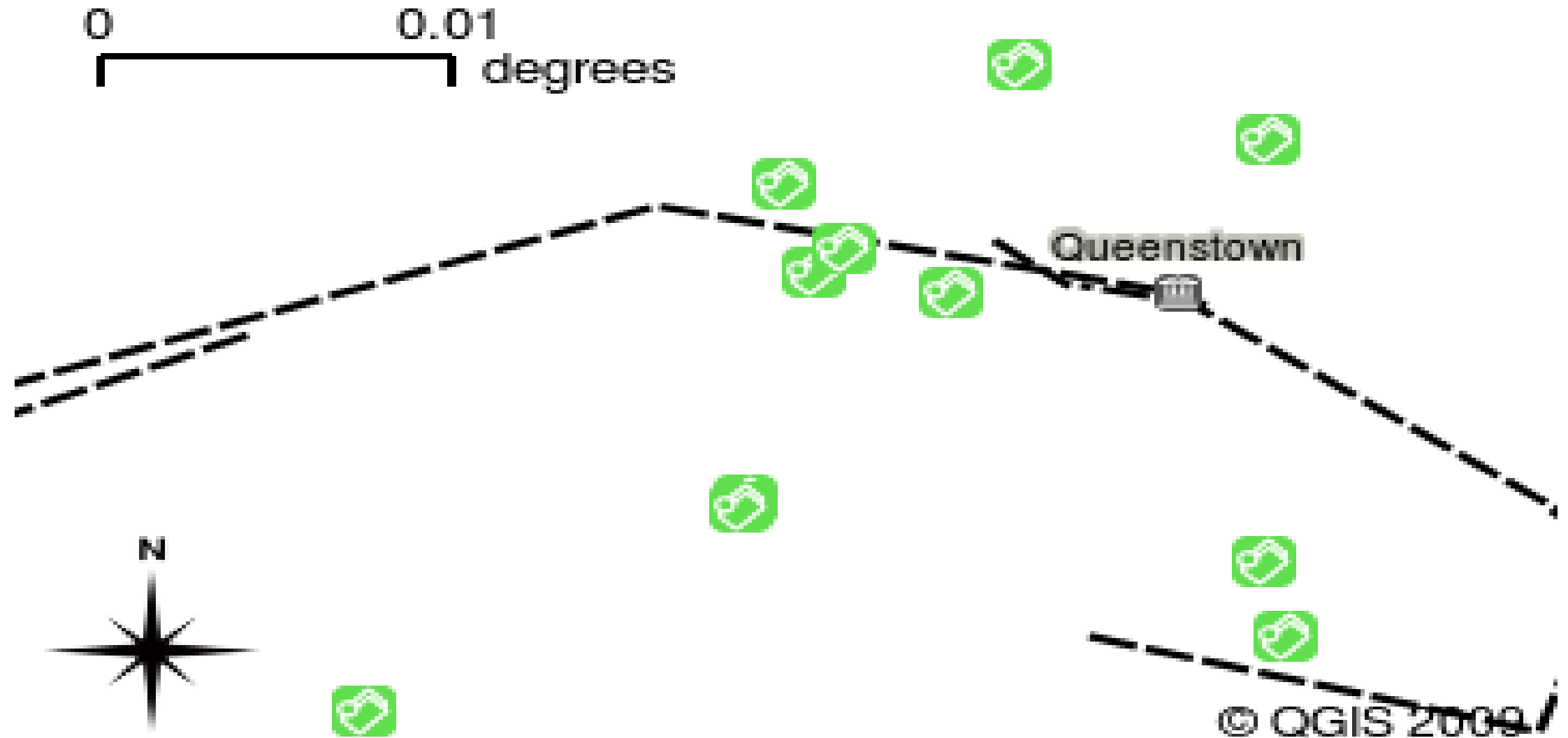
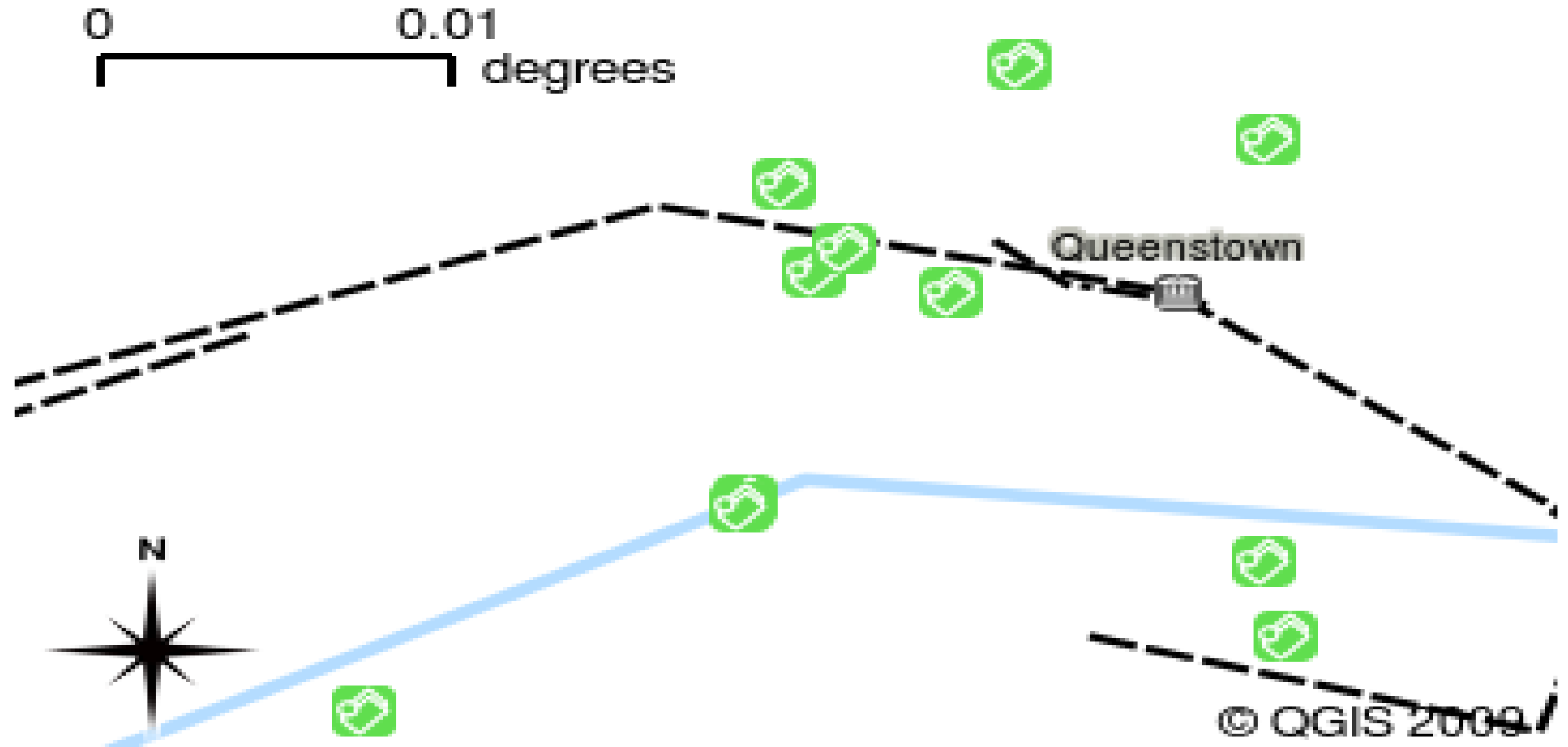


Fig. 6 A map view with several layers being added to it. A rivers layer added to the map view



Map legend.

- The map legend provides a list of layers that have been loaded in the GIS Application.
- Unlike a paper map legend, the map legend or 'layers list' in the GIS Application provides a way to
 - re-order, hide,
 - show and group layers.
- Changing the layer order is done by clicking on a layer in the legend, holding the mouse button down and then dragging the layer to a new position.
- In figures Fig. 5 and Fig. 6 the map legend is shown as the area to the left of the GIS Application window.
- By changing the layer order, the way that layers are drawn can be adjusted —
- in this case so that rivers are drawn below the roads instead of over them.

Getting a GIS Application for your own computer(s)

- There are many different GIS Applications available.
- Some have many sophisticated features and cost tens of thousands of Rands for each copy(e.g. ArcGIS).
- In other cases, you can obtain a GIS Application for free.
- Deciding which GIS Application to use is a question of how much money you can afford and personal preference.
- we will be using the QGIS/ArcGIS Application(s).
- QGIS is completely free and you can copy it and share it with your friends as much as you like. ArcGIS is Costly(but crack versions are available freely)
- visit <https://www.qgis.org/> to download your free copy if you have access to the internet.

GIS Data

- Now that we know what a GIS is and what a GIS Application can do, let's talk about **GIS data**.
- Data is another word for **information**.
- The information we use in a GIS normally has a **geographical aspect** to it.
- Think of our example above, about the health care worker. who created a table to record diseases that looked like this:

Longitude	Latitude	Disease	Date
26.870436	-31.909519	Mumps	13/12/2008

- The longitude and latitude columns hold **geographical data**.
- The disease and date columns hold **non-geographical data**.

GIS Data

- GIS Systems work with many different types of data.
- **Vector data** is stored as a series of X, Y coordinate pairs inside the computer's memory.
- Vector data is used to represent points, lines and areas.
- Illustration Fig. 5(on Next Slide), shows different types of vector data being viewed in a GIS application.
- In this lecture ,we will be exploring vector data in more detail

GIS Data

Vector data is used to represent

- *points (e.g. towns),*
- *lines (e.g. rivers) and*
- *polygons (e.g. municipal boundaries).*



Fig. 6

GIS Data

- are stored as a grid of values.
- There are many satellites circling the earth and the photographs they take are a kind of raster data that can be viewed in a GIS.
- One important difference between raster and vector data is that if you zoom in too much on a raster image, it will start to appear 'blocky' (see illustrations Fig. 7 and Fig. 8).
- In fact these blocks are the individual cells of the data grid that makes up the raster image.
- We will be looking at raster data in greater detail later.

Raster data



Fig. 7 *Raster data are often images taken by satellites. Here we can see mountains in the Eastern Cape*



Fig. 8 *The same raster data, but this time zoomed in. The grid nature of the data can be seen*

What have we learned?

- Let's wrap up what we covered in this Lecture:
- A **GIS** is a system of computer hardware, computer software and geographical data.
- A **GIS Application** allows you to view geographical data and is an important part of the GIS.
- A GIS Application normally consists of a **menu bar**, **toolbars**, a **map view** and a **legend**.
- **Vector** and **raster** data are geographical data used in a GIS application.
- **Geographical** data can have associated **non-geographical** data.

Now you try!

- Here are some ideas for you to try with your learners:
- **Geography:** Describe the concept of GIS as outlined in this Lecture. Please try to think of 3 reasons why it might be handy to use a GIS instead of paper maps.

- Here are some that we could think of:
 - GIS Applications allow you to create many different types of maps from the same data.
 - GIS is a great visualization tool that can show you things about your data and how they are related in space (e.g. those disease outbreaks we saw earlier).
 - Paper maps need to be filed and are time consuming to view. The GIS can hold a very large amount of map data and make it quick and easy to find a place you are interested in.

- **Something to think about**
- If you don't have a computer available, many of the topics we cover in this tutorial can be reproduced using an overhead and transparency as it uses the same technique of layering information. However, to properly understand GIS it is always better to learn it using a computer.
- **Further reading**
- **Book:** Desktop GIS: Mapping the Planet with Open Source Tools. **Author:** Gary Sherman. **ISBN:** 9781934356067
- **What's next?**
- In the sections that follow we are going to go into more detail, showing you how to use a GIS Application.
- Next up, let's look at vectors!